

## ASSIGNMENT N°1 OF CALCULUS FOR ENGINEERS

### Group 1.

**Question1.** Sketch the graph of the functions:

$$\text{a) } h(x) = \begin{cases} x, & 0 < x \leq 1 \\ 3 - x, & 1 < x \leq 2 \\ 5, & 2 < x \leq 3 \\ 0, & \text{elsewhwre} \end{cases} \quad \text{b) } f(x) = \begin{cases} x^2 + 1, & x > 1 \\ 2x, & x \leq 1 \end{cases}$$

**Question2.** Determine the domain of function and the inverse function and the existence condition of those inverse functions of the following functions:

$$\text{a) } y = -3 + \ln 2x \quad \text{c) } y = \sqrt{2 + x - x^2}$$

$$\text{b) } y = \sqrt[3]{x+1} \quad \text{d) } y = \frac{1}{4-x^2}$$

**Question3.** The following functions are given as composite functions, then write them as un composite functions i.e  $y = f(x)$

$$\text{a) } y = u^2 ; \quad u = \sin x \quad \text{b) } y = \arctan u ; \quad u = \sqrt{v} ; v = \log 2x$$

$$\text{c) } y = \begin{cases} u^2 + 1, & u > 1 \\ 2u, & u \leq 1 \end{cases} \quad u = x^2 + 2x - 4$$

**Question4.** Find the explicitly function of function  $y$  given by the equations:

$$1. \quad x^2 - \arccos y = \pi$$

$$2. \quad 10^x + 10^y = 10$$

$$3. \quad x + |y| = 2y$$

**Question5.** Evaluate the following limits:

$$\text{a) } \lim_{x \rightarrow \infty} \left( \frac{x^2 - 2x + 3}{x^2 - 3x + 2} \right)^{x+2} \quad \text{c) } \lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{x}$$

$$\text{b) } \lim_{x \rightarrow +2} \left( \frac{\sqrt{x^2 + x + 3} - (x+1)}{x^2 - 4} \right) \quad \text{d) } \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\sqrt[3]{x+1} - 1}$$

**Question6.** Solve the following equation and system of equation:

$$\text{a) } \log_3(4 \cdot 3^x - 1) = 2x + 1 \quad \text{b) } \begin{cases} \ln(x-2) - 3 \ln(y-1) = 9 \\ 2 \ln(x-2) - \ln(y-1) = 4 \end{cases} \quad \text{c) } \ln(x-2) < 3$$

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### Group 2.

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{-x} + \frac{1}{\sqrt{2+x}}$       c)  $y = \log_{2x-1}\left(\frac{x^2-3x+2}{x+1}\right)$   
b)  $y = \sqrt{\sin - 2x}$       d)  $y = \sqrt[4]{\frac{2x+1}{\sqrt{1-2x}}}$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = x^2 - 1$   
b)  $y = \log(x/2)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x}}$       c)  $\lim_{x \rightarrow \pi} \frac{1 - \sin x/2}{\pi - x}$   
b)  $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{2x}{x+1}}$       d)  $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x}-1}{\sqrt[4]{x}-1}$

**Question4.**

- a) Calculate the derivative of the following function  $y = \frac{x}{a^2\sqrt{a^2+x^2}}$   
b) Find the slope of the tangent line at (4, 0) to the graph of  $7y^4 + x^3y + x = 4$   
c) For the function  $f(x) = 4x^3 - 3x^2 - 6x + 12$  determine the maximum, minimum and inflection points

**Question5.** Calculate the derivative of the following functions

a)  $y = \tan^5(5x)$   
b)  $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$   
c) Compute  $\frac{dy}{dx}$  for  $t = 1$ , if  $\begin{cases} x = t \ln t \\ y = \frac{\ln t}{t} \end{cases}$

**Question6.** a) Show that:  $\frac{1}{\log_a x} + \frac{1}{\log_b x} = \frac{1}{\log_{ab} x}$ ,  $\forall a, b \in \mathbb{R}_0^+ \setminus \{1\}, \forall x > 0$

b) Solve the following equation and system of equation: i)  $4^x - 3^{x-1/2} = 3^{x+1/2} - 2^{2x}$  ii)

$\begin{cases} x^2 + y^2 = 130 \\ \ln(-x) + \ln(-y) = \ln 63 \end{cases}$       iii)  $\ln(2x + 3) \leq 3$

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### Group 3.

#### Question1.

- a) Determine the domain of definition of the following functions:  $y = \sqrt{2 - x - x^2}$
- b) Sketch the graph of this function:  $y = |x - 2| + |2x - 6| + |-x - 1|$  in  $[-5, 5]$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

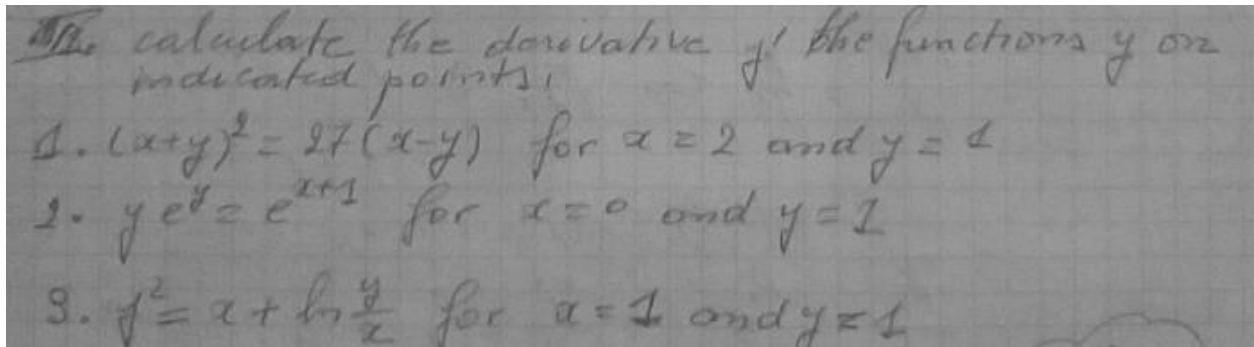
- a)  $y = 5e^{4x}$
- b)  $y = \sec^{-1}(3x)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} \left( \frac{-5x^2 - 2x + 3}{x^2 - 3x + 2} \right)^{\frac{\sin x}{x}}$       b)  $\lim_{x \rightarrow -\infty} \left( \frac{x}{\sqrt{-x^2 - 3x + 2}} \right)$

**Question4.a)** Locate the relative extrema of  $f(x) = 3x^{5/3} - 15x^{2/3}$

b) On which intervals is the function  $f(x) = x^3 - 3x^2 + 1$  increasing? Decreasing?



**Question5.** Show that:

a)  $\log_a x \cdot \log_b x + \log_b x \cdot \log_c x + \log_a x \cdot \log_c x = \frac{\log_a x \cdot \log_b x \cdot \log_c x}{\log_{abc} x}, \quad \forall a, b, c \in \mathbb{R}_0^+ \setminus \{1\}, \forall x > 0$

b)  $\frac{e^x - 1}{e^x + 1} = \frac{1 - e^{-x}}{1 + e^{-x}}$

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### Group 4.

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{-2x + 3} + \frac{1}{\sqrt{2-x}}$

b)  $y = \sqrt{\sin(-2x)}$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = x^2 + 5x + 2$

b)  $y = \log\left(\frac{2x+1}{3}\right)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} (1 - \cos x)^{\frac{1}{x}}$

b)  $\lim_{x \rightarrow \infty} \left(\frac{x}{2x+7}\right)^{\frac{2x}{x+1}}$

**Question4.** Calculate the derivative of the following functions

a)  $y = \frac{x}{a^2 \sqrt{a^2 + \frac{1}{x}}}$

b)  $y = \sec^{-1}\left(\frac{x^2+1}{x^2}\right)$

**Question5.** Consider the function defined by  $y = x^2 + 1$  and the line  $D \equiv x + y = 3$  on the interval  $[0, 2]$ .

a) Find the intersection points between the line and the curve of the function.

b) Draw the function and the line in the same plane

**Question6.** Show that:

a)  $\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} + \dots + \frac{1}{\log_n x} = \frac{1}{\log_{n!} x}, \quad \forall x > 0$

b)  $e^{-x} - e^{-2x} = \frac{e^x - 1}{e^{2x}}$

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### Group 5.

**Question1.** Determine the domain of definition of

a)  $f(x) = \log\left(\frac{2+x}{2-x}\right)$

b)  $g(x) = \frac{\ln 2-x}{\sqrt{x-4}}$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = -2 + 6e^{4x}$

b)  $y = \cotan(3x)$

**Question3.** Calculate the limits of

a)  $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2-x+1}}{3x-2}$

b)  $\lim_{x \rightarrow 0} \frac{3x}{\tan 2x}$

**Question4.** Calculate the derivative of the following functions

a)  $y = \tan^5(-5x)$

b)  $y = \frac{\cos^{-1} x}{\sqrt{1-x^2}}$

**Question5.** Given function  $f(x) = \frac{x+3}{x-2}$

- a) Find domain of definition and boundary limits
- b) Find asymptotes to the curve
- c) Compute the first derivative and study its sign
- d) Compute the second derivatives and study its sign
- e) Find intercept points if any
- f) Sketch the graph.

**Question6.** a) Calculate

i)  $\frac{1}{16}^{-2 \log_4 12}$     ii)  $2^{\log_{2\sqrt{2}} 15}$

b) Determine the basis 'a' if we have:

i)  $\log_a 7 = 0.7$     ii)  $\log_a 0.1 = 2$     iii)  $\log_a 16 = 2$     iv)  $\log_a 2 = -2$

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### Group 6.

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{2 + x - x^2}$

b)  $y = \cos^{-1}\left(\frac{2x}{1+x}\right)$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = -3 + e^{4x}$

b)  $y = \tan^{-1}(3x)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} \left( \frac{x^2 - 2x + 3}{x^2 - 3x + 2} \right)^{\frac{\sin x}{x}}$

b)  $\lim_{x \rightarrow +\infty} \left( \frac{x}{\sqrt{x^2 - 3x + 2}} \right)$

**Question4.** Calculate the derivative of the following functions

a)  $y = \tan^5(5x)$

b)  $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$

**Question5.** a) Calculate

i)  $\frac{1}{16}^{-2 \log_4 12}$       ii)  $2^{\log_{2\sqrt{2}} 15}$

b) Show that:

i)  $\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} + \dots + \frac{1}{\log_n x} = \log_x n!, \quad \forall x \in \mathbb{R}^+ \setminus \{0,1\}$

ii)  $\log_a x \cdot \log_b x + \log_b x \cdot \log_c x + \log_a x \cdot \log_c x = \frac{\log_x abc}{\log_x a \cdot \log_x b \cdot \log_x c}, \quad \forall a, b, c \in \mathbb{R}_0^+ \setminus \{1\}, \forall x \in \mathbb{R}_0^+ \setminus \{1\}$

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### Group 7.

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{-x} + \frac{1}{\sqrt{2+x}}$       b)  $y = \sqrt{\sin 2x}$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of: a)  $y = x^2 - 1$       b)  $y = \log(x/2)$

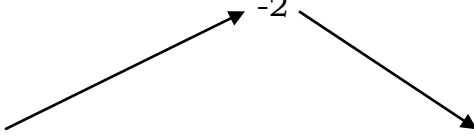
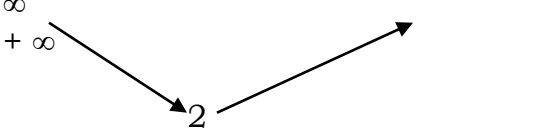
**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x}}$       b)  $\lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^{\frac{2x}{x+1}}$

**Question4.** Calculate the derivative of the following functions

a)  $y = \frac{x}{a^2 \sqrt{a^2 + x^2}}$       b)  $y = \sin^{-1} \left( \frac{x^2 - 1}{x^2} \right)$

**Question5.** Consider a function  $f$  whose the following variation table:

| $x$     | $-\infty$<br>$+\infty$                                                              | $-2$ |     |     | $-1$ | $0$                                                                                  |     |     |     |     |     |
|---------|-------------------------------------------------------------------------------------|------|-----|-----|------|--------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|
| $f'(x)$ | $+$                                                                                 | $+$  | $0$ | $-$ | $-$  | $-$                                                                                  | $-$ | $0$ | $+$ | $+$ | $+$ |
| $f(x)$  |  |      |     |     |      |  |     |     |     |     |     |

Then let  $f$  be a function of the form  $f(x) = ax + b + \frac{c}{x+1}$ , where  $a, b$  and  $c$  are the real numbers:

- Find the absolute maximum and absolute minimum
- Calculate  $f'(x)$  and  $f''(x)$
- Use the data from above table to find the real coefficients  $a, b$  and  $c$ .
- Determine different types of asymptotes and their equations.

**Question6.** If  $\ln 2 = 0.6931$ ,  $\ln 3 = 1.0986$  and  $\ln 5 = 1.6094$ , then calculate

- a)  $\ln 10$    b)  $\log_2 3$    c)  $\ln 30$    d)  $\ln 7.5$    e)  $\log_2 10$

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### Group 8.

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{2 - x - x^2}$

b)  $y = \cos^{-1}\left(\frac{-2x}{1+x}\right)$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = 5e^{4x}$

b)  $y = \sec^{-1}(3x)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} \left( \frac{-5x^2 - 2x + 3}{x^2 - 3x + 2} \right)^{\frac{\sin x}{x}}$

b)  $\lim_{x \rightarrow -\infty} \left( \frac{x}{\sqrt{-x^2 - 3x + 2}} \right)$

**Question4.**

1. Calculate the derivative of the following functions

a)  $y = \sec^3 5x$     c)  $y = e^{\frac{1}{1-x}}$

b)  $y = 7^{2x-7}$     d)  $y = \frac{\sec^{-1} 5x}{\sqrt{1+x^2}}$

2. A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 14 m by 22 m by cutting out equal squares of side  $x$  at each corner and then folding up the sides. Express the volume  $V$  of the box as a function of  $x$  and find the value of  $x$  for which the volume is maximum

**Question5.** Calculate the derivatives  $y'_x$  of the following expressions:

1.  $\begin{cases} y = a(\cos t + t \sin 2t) \\ x = a(\cos \sqrt{t} - \sin t^2) \end{cases}$     2.  $\begin{cases} y = (2t + 3)e^{-t} \\ x = \frac{e^{2t+3}}{2-t} \end{cases}$

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### Group 9

**Question1.** Determine the domain of definition of the following functions:

a)  $y = \sqrt{-2x + 3} + \frac{1}{\sqrt{2-x}}$

b)  $y = \sqrt{\sin(-2x)}$       c)  $y = \frac{\sqrt{x^2-4}}{x^2-8x+15}$  .

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = x^2 + 5x + 2$

b)  $y = \log\left(\frac{2x+1}{3}\right)$

**Question3.** Evaluate the following limits:

a)  $\lim_{x \rightarrow 0} (1 - \cos x)^{\frac{1}{x}}$       b)  $\lim_{n \rightarrow +\infty} \left( \frac{1}{n^2} + \frac{2}{n^2} + \frac{3}{n^2} + \dots + \frac{n-1}{n^2} \right)$

c)  $\lim_{n \rightarrow +\infty} \left[ \frac{1+3+5+7+\dots+(2n-1)}{n+1} - \frac{2n+1}{2} \right]$       d)  $\lim_{x \rightarrow \infty} \left( \frac{x}{2x+7} \right)^{\frac{2x}{x+1}}$

**Question4.** Calculate the derivative of the following functions

a)  $y = \frac{x}{a^2 \sqrt{a^2 + \frac{1}{x}}}$

b)  $y = \sec^{-1} \left( \frac{x^2+1}{x^2} \right)$

c) Find two numbers whose sum is 20 and whose product is:

i) maximum      ii) minimum

**Question5.** Calculate the derivatives  $y'_x$  of the following expressions:

1.  $\begin{cases} y = a(\ln \tan \frac{t}{2} + \cos t - \sin 2t) \\ x = a(\cos \sqrt{t+5} - \sin 3t^2) \end{cases}$       2.  $\begin{cases} y = (2t+3)e^{-t} \\ x = \frac{e^{2t+3}}{2-t} \end{cases} \text{ at } t = 0$

3.  $\begin{cases} y = e^t \cos t \\ x = e^t \sin 2t \end{cases} \text{ at } t = \frac{\pi}{4}$

## ASSIGNMENT N°1 OF CALCULUS FOR ENGINEERS

### Group 10

**Question1.** Simplify

a)  $\sin x \cos x \tan x + \frac{2 \sin x \cos^3 x}{\sin 2x}$

b)  $\tan x + \frac{1}{\tan x}$

**Question2.** Determine the inverse function and the existence condition of those inverse functions of:

a)  $y = -2 + 6e^{4x}$

b)  $y = \cotan^{-1}(3x)$

**Question3.** Solve the following trigonometric equations:

a)  $5 \cos(3x) + 4 = 0; \forall x \in [0, 2\pi]$

b)  $\tan\left(\frac{3x+30^\circ}{4}\right) = -1.6319; -270^\circ \leq x \leq 270^\circ$

**Question4.** Calculate the derivative of the following functions

a)  $y = \tan^5(-5x)$

b)  $y = \frac{\cos^{-1} x}{\sqrt{1-x^2}}$

**Question5.**

a) Find the dimensions of a rectangle with perimeter 100m whose area is as large as possible?

b) Find the slope of the tangent line at  $(\pi, \frac{\pi}{2})$  to the graph of  $5y^2 + \sin y = x^2$

c) Verify that the hypotheses of Mean value theorem are satisfied on the given interval and find all values of  $c$  that satisfy the conclusion of the theorem:

1)  $f(x) = x^2 + x; [-4, 6]$

2)  $f(x) = x^3 + x - 6; [-1, 24]$